

Yaze (Arthur) Li

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EDUCATION

•University of Arkansas

PhD, Electrical Engineering

August 2023

Fayetteville, AR

Thesis: Achieving High Renewable Energy Integration in Smart Grids with Machine Learning

•Tsinghua University

BS, Electrical, Electronics and Communication Engineering

July 2017

Beijing, China

SKILLS

- **Programming Languages:** Python, MATLAB, R, C, MySQL
- **Machine Learning Frameworks:** Tensorflow, PyTorch, NumPy, Pandas, Stable Baselines
- **Engineering Tools:** Simulink, PSS/E, TARA, ASPEN OneLiner, OPAL-RT, LabVIEW, L^AT_EX, AMPL
- **Electrical Engineering Skills:** Power flow, State estimation, Short circuit analysis, SCADA system

EXPERIENCE

•Power system engineer

CF Power Ltd. - an E Source Company

August 2024 - Now

Calgary, AB

- Perform Transmission Full Interconnection Study for AEP projects in ERCOT using PSS/E, TARA, ASPEN OneLiner and other softwares, including power flow, voltage stability, transient stability, voltage ride-through, and short circuit analysis.
- Perform Do-No-Harm study for projects in PJM using TARA; Perform PJM Cluster Study including data review, model development, case build, writing scope of work, and transient stability study.
- Develop PSS/E automation tool and perform transient stability study for AESO Cluster Study.

•Graduate Assistant

Cybersecurity Center for Secure Evolvable Energy Delivery Systems (SEEDS), University of Arkansas

August 2017 - July 2023

Fayetteville, AR

- Developed a Coordinate Descent (CD) based dynamic programming (DP) approach for optimizing battery-assisted PV systems, reducing total cost by 29.3% over ten years using MATLAB. [Journal][GitHub]
- Implemented an online scheduling algorithm for battery-assisted PV systems using reinforcement learning (DDPG), achieving a 3.29% annual cost reduction with MATLAB and Stable Baselines. [Journal][GitHub]
- Proposed a DDPG algorithm for scheduling optimal real-time power flow (RT-OPF) in a microgrid equipped with energy storage and power generations, using Matpower in MATLAB and Stable Baselines. [GitHub]
- Developed a new framework and algorithm of truck-based mobile energy couriers (MEC) scheduling in a distribution network with renewable energy sources, using Pandapower and Stable Baselines in Python. [Journal][GitHub]

•Graduate Assistant

Center for Infrastructure Trustworthiness in Energy Systems (CITES), University of Arkansas

August 2019 - July 2023

Fayetteville, AR

- Proposed a dynamic watermarking-based active low latency attack detection algorithm for grid-connected PV systems, achieving a detection delay of 50 ms with PFA below 5% in MATLAB. [Journal][GitHub]
- A Deep Q-Network (DQN) based deep reinforcement learning algorithm has been proposed for the low latency detection of cyberattacks, such as FDI and DoS attacks in smart grids, using Power system Toolbox in MATLAB and Stable Baselines. [Journal][GitHub]

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- Y. Li** and J. Wu, “Optimum design of battery-assisted photo-voltaic energy system for a commercial application,” in Power & Energy Society General Meeting (PESGM). IEEE, 2019. [Conference][GitHub]
 - Y. Li** and J. Wu, “Optimum integration of solar energy with battery energy storage systems,” IEEE Transactions on Engineering Management, 2020. [Journal][GitHub]
 - S. Nath, **Y. Li**, J. Wu, and P. Fan, “Multi-user multi-channel computation offloading and resource allocation for mobile edge computing,” in ICC 2020-2020 IEEE International Conference on Communications (ICC). IEEE, 2020, pp. 1–6. [Conference]
 - Y. Li** and J. Wu, “Low latency cyberattack detection in smart grids with deep reinforcement learning,” International Journal of Electrical Power & Energy Systems, vol. 142, p. 108265, 2022. [Journal][GitHub]
 - Y. Li**, N. Lin, J. Wu, Y. Pan and Y. Zhao, "Low Latency Attack Detection with Dynamic Watermarking for Grid-Connected Photovoltaic Systems," in IEEE Journal of Emerging and Selected Topics in Industrial Electronics, doi: 10.1109/JESTIE.2023.3327051 [Journal][GitHub]
 - Y. Li**, J. Wu and Y. Pan, "Deep Reinforcement Learning for Online Scheduling of Photovoltaic Systems with Battery Energy Storage Systems," in Intelligent and Converged Networks, vol. 5, no. 1, pp. 28-41, March 2024, doi: 10.23919/ICN.2024.0003 [Journal][GitHub]
 - Y. Li**, J. Wu and Y. Pan, “Optimum Scheduling of Truck-Based Mobile Energy Couriers (MEC) Using Deep Deterministic Policy Gradient,” in Intelligent and Converged Networks, vol. 6, no. 3, pp. 195-208, September 2025, doi: 10.23919/ICN.2025.0016 [Journal][GitHub]

RESEARCH INTEREST

Power System analysis, Detection and Estimation, Renewable Energy Systems